

SOLUTION REQUIREMENTS

Script-Controlled ACL District Record Access in ServiceNow

The Script-Controlled ACL District Record Access in ServiceNow project is designed to provide a secure, role-based mechanism for controlling access to district records. In organizations that manage information across multiple districts, unauthorized access can lead to security breaches, data leakage, and compliance issues. The proposed solution leverages ServiceNow Access Control Lists (ACLs) and scripting mechanisms to ensure that only authorized users can access records belonging to their assigned district.

The system performs user authentication, role verification, district validation, and ACL execution before granting access. This approach improves security, reduces manual administration, and supports organizational governance requirements.

Project Objectives

- Provide district-level record security.
- Restrict unauthorized access to sensitive information.
- Automate access validation using ACL scripts.
- Reduce administrative effort in permission management.
- Maintain compliance with security policies.

Functional Requirements

FR-1 User Authentication

The system shall authenticate all users before access is granted.

FR-2 Role Verification

The system shall validate user roles before executing operations.

FR-3 District Validation

The system shall compare the user's district with the record district.

FR-4 Read Access

Authorized users can view only district-specific records.

FR-5 Create Access

Authorized users can create records within their district.

FR-6 Update Access

Authorized users can modify permitted records.

FR-7 Delete Access

Authorized users can delete records if permitted.

FR-8 Administrative Access

Administrators have complete system access.

FR-9 Audit Logging

All security events and access attempts are logged.

Non-Functional Requirements

Security

The solution must enforce strict district-based access restrictions and prevent unauthorized data exposure.

Performance

Validation should be completed quickly without noticeable delay.

Reliability

ACL decisions must remain consistent under all conditions.

Scalability

The system should support a growing number of districts, users, and records.

Availability

Security controls should be available whenever the platform is operational.

Maintainability

ACL scripts and role definitions should be easy to update.

Hardware and Software Requirements

Hardware	Specification
Processor	Intel i3 / Ryzen 3 or higher
RAM	8 GB Minimum
Storage	256 GB SSD or higher
Internet	Stable Connection

Software	Purpose
ServiceNow	Platform Development
JavaScript	ACL Scripting
Web Browser	System Access
Windows/Linux	Development Environment

Constraints

The system depends on accurate district assignments, properly configured user roles, and valid ACL scripts. Any misconfiguration may affect access-control decisions. Security policies must always take precedence over convenience.

Proposed Solution and Benefits

The proposed solution utilizes Script-Controlled ACLs in ServiceNow to enforce district-level security. Whenever a user requests access to a record, the system validates authentication credentials, verifies roles, executes ACL scripts, and compares district information. Access is granted only when all validation checks succeed. This approach ensures record-level security while minimizing manual administrative effort.

Benefits

- Improved district-level security
- Protection of sensitive records
- Automated access-control decisions
- Reduced administrative workload
- Better auditing and compliance
- Scalable and maintainable security architecture